

INFINITE COMPUTER SOLUTIONS — FINAL ROUND HACKATHON 2026

MUGEN AI: An Autonomous Policy-Aware Asset Request Agent

Use Case & System Diagrams Report

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1 System Actors & Use Cases

1.1 Actors

Actor	Description
Employee (User)	Any staff member who initiates an asset request via Telegram.
Admin	Privileged user who uploads rulebook PDFs and receives security alerts.
HRIS System	Mock JSON data store providing employee grade, department, and status.
Decision Engine	Internal LLM agent that evaluates eligibility and produces verdicts.
Suspicion Scorer	Internal security module that scores every incoming message.
ChromaDB (RAG)	Vector store queried during policy validation.

Table 1: System actors and their roles in MUGEN AI.

1.2 Use Case Summary

UC#	Use Case	Primary Actor	Brief Description
UC1	Submit Asset Request	Employee	Multi-turn slot-filling to capture asset type, model, justification, and urgency.
UC2	Upload Rulebook PDF	Admin	Admin sends PDF; bot ingests, chunks, embeds, and stores in ChromaDB.
UC3	Validate Employee Data	Decision Engine	Fetch grade, department, and active status from HRIS.
UC4	Retrieve Policy Context	Decision Engine	Embed query and retrieve top-3 rulebook chunks from ChromaDB.
UC5	Approve / Reject / Flag	Decision Engine	LLM reasons step-by-step and outputs structured verdict with policy reference.
UC6	Detect Suspicious Input	Suspicion Scorer	Score for injection patterns, grade mismatch, velocity, and after-hours access.
UC7	View Request Status	Employee	<code>/status <id></code> queries the SQLite <code>asset_requests</code> table.
UC8	View Request History	Employee	<code>/history</code> returns the last 5 requests for the employee.
UC9	Receive Admin Alert	Admin	Flagged sessions trigger a real-time Telegram DM to the admin.

Table 2: Complete use case catalogue for MUGEN AI.

2 Use Case Diagram

2.1 Description

The use case diagram shows the interactions between external actors (Employee and Admin) and all nine use cases within the MUGEN AI system boundary, including *includes* and *extends* relationships.

2.2 Diagram

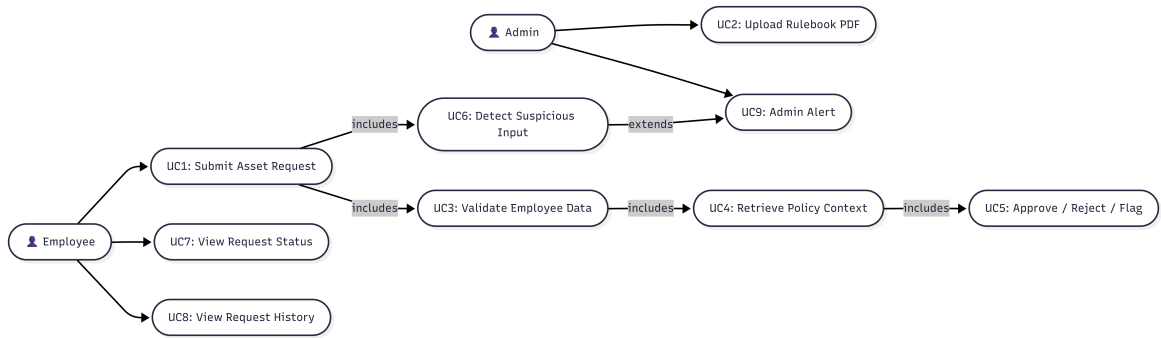


Figure 1: Use Case Diagram — MUGEN AI. All nine use cases mapped to their primary actors with *includes* and *extends* relationships.

3 Sequence Diagram

3.1 Description

The sequence diagram captures the full message lifecycle for a standard asset request — from the initial `/request` command through slot extraction, suspicion scoring, HRIS validation, RAG retrieval, LLM decision, database persistence, and the final confirmation card delivered to the user.

3.2 Diagram

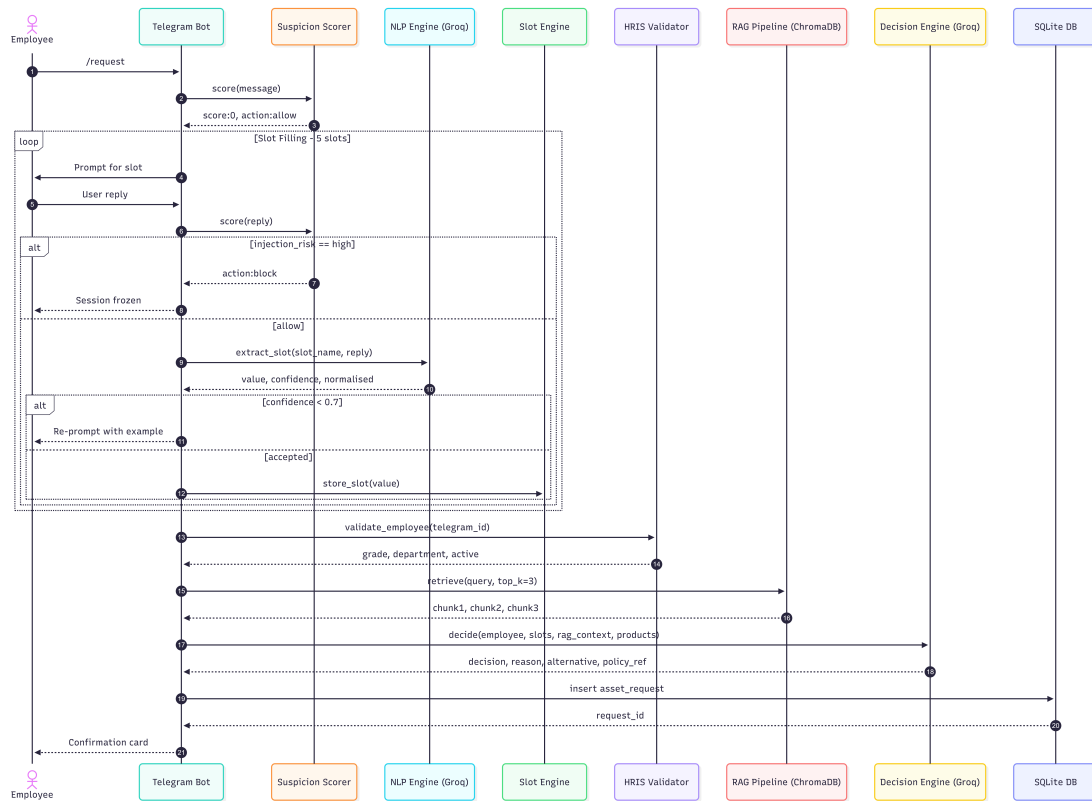


Figure 2: Sequence Diagram — Asset Request Lifecycle. Full message flow from Telegram user through NLP engine, suspicion scorer, slot engine, HRIS, RAG pipeline, decision engine, and SQLite persistence.

4 Activity / Flowchart Diagram

4.1 Description

The activity diagram models the complete decision logic as a flowchart — starting at message receipt, through the suspicion gate, iterative slot-filling, HRIS validation, RAG-augmented policy checking, and the three-branch decision outcome: Approved, Flagged, or Rejected.

4.2 Diagram

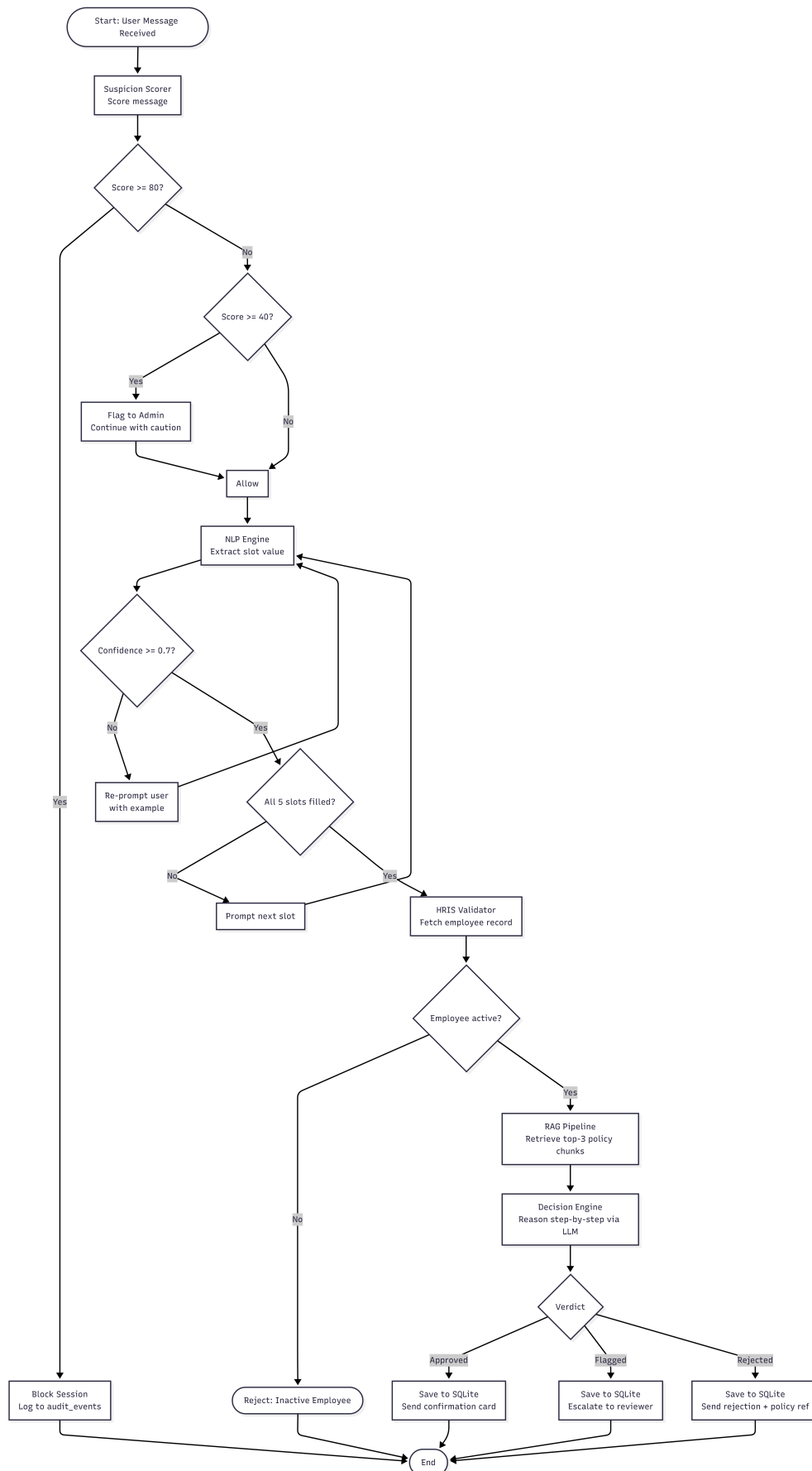


Figure 3: Activity / Flowchart Diagram — MUGEN AI Decision Logic.

Full decision flow from message receipt through suspicion gating, slot extraction, HRIS validation, RAG-augmented policy evaluation, and three possible verdicts.

5 System Architecture Diagram

5.1 Description

The architecture diagram illustrates the seven-layer system design of MUGEN AI: Telegram interface, NLP engine, slot-filling engine, PDF/RAG pipeline, HRIS validation, LLM decision engine, and SQLite persistence — with the suspicion detection layer running horizontally across all layers.

5.2 Diagram

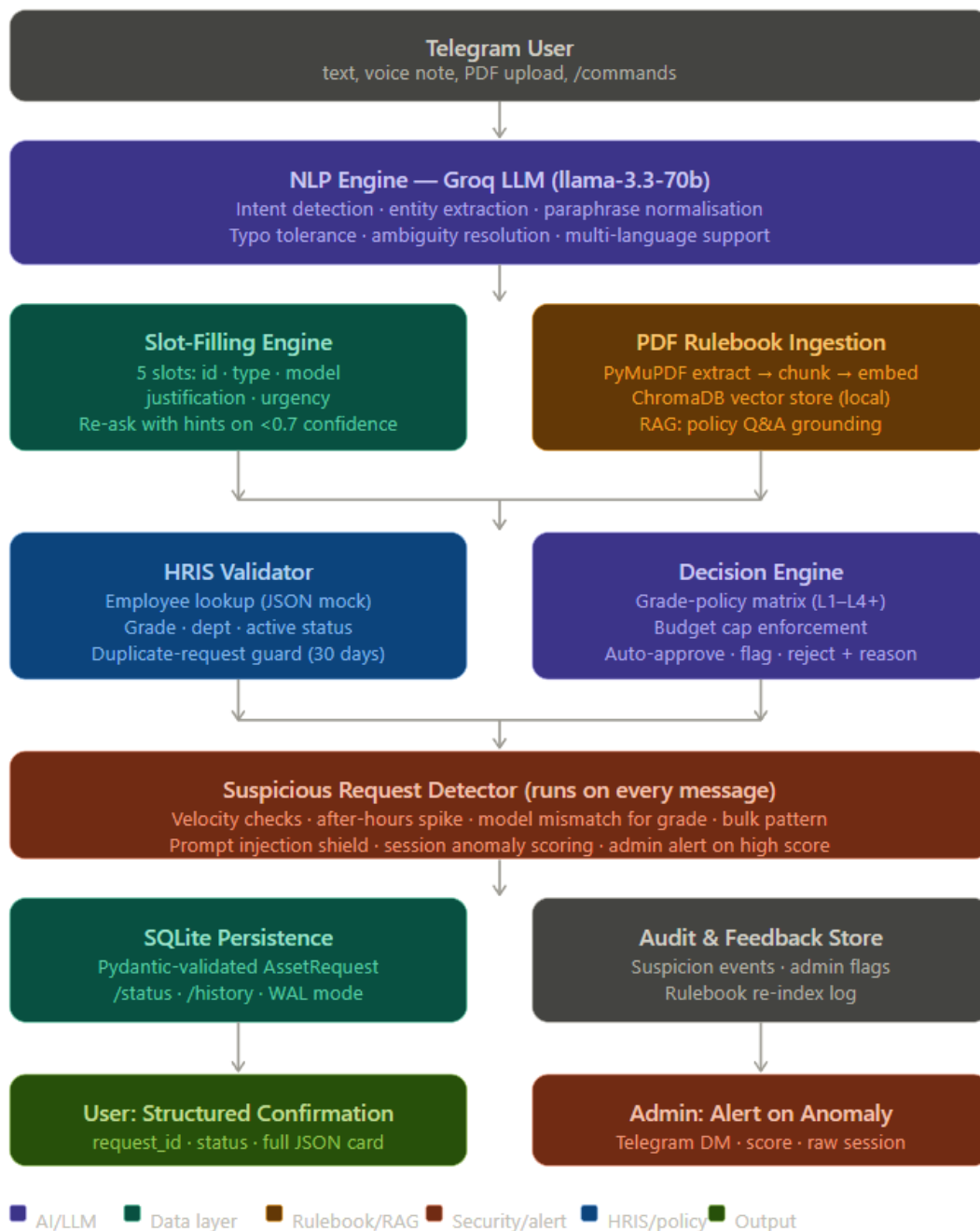


Figure 4: MUGEN AI — Full System Architecture. Seven horizontal layers: (1) Telegram Interface, (2) NLP Engine, (3) Slot-Filling Engine, (4) PDF Rulebook Ingestion & RAG Pipeline, (5) HRIS Validator & Decision Engine, (6) Suspicion Detection & Admin Alerts, (7) SQLite Persistence. The suspicion scorer intercepts every message before any business logic executes.

5.3 Layer Summary

#	Layer	Key Components
1	Telegram Interface	python-telegram-bot v20; /start, /request, /upload_rulebook, /status, /history
2	NLP Engine	Groq API (llama-3.3-70b-versatile); LangChain LCEL; slot extractor with injection risk scoring
3	Slot-Filling Engine	ConversationHandler; 5-state FSM; Pydantic v2 models; 10-min session timeout
4	RAG Pipeline	PyMuPDF; RecursiveCharacterTextSplitter (400 tok / 60 overlap); all-MiniLM-L6-v2; ChromaDB
5	HRIS & Decision	hris.json; asset_policy.json; products.json; Groq decision prompt with structured JSON output
6	Suspicion Detection	Python scoring class; regex patterns; after-hours / velocity / grade-mismatch flags; admin DM
7	Persistence	SQLite (WAL mode); aiosqlite; asset_requests table; audit_events table

Table 3: Seven architectural layers of MUGEN AI and their key components.

6 Use Case Specifications

6.1 UC1 — Submit Asset Request

Use Case ID	UC1
Name	Submit Asset Request
Primary Actor	Employee
Precondition	Employee has a valid Telegram account and sends /request.
Main Flow	Bot prompts for five slots: Employee ID, asset type, model, justification, urgency. NLP extracts each slot; confidence < 0.7 triggers re-prompt. All slots are passed to the decision engine.
Alt Flow 1	Employee sends /cancel — session reset, event logged.
Alt Flow 2	Session inactive 10 min — auto-timeout, event logged.
Alt Flow 3	Employee sends image or sticker — graceful redirect message.
Exception	Suspicion score ≥ 80 — session blocked; audit event created.
Postcondition	Request stored in <code>asset_requests</code> with verdict; confirmation card sent.

6.2 UC2 — Upload Rulebook PDF

Use Case ID	UC2
Name	Upload Rulebook PDF
Primary Actor	Admin
Precondition	Sender Telegram ID is in the admin whitelist in <code>.env</code> .
Main Flow	Admin sends PDF to bot. Bot extracts text via PyMuPDF, splits into 400-token chunks (60-token overlap), embeds via all-MiniLM-L6-v2, and stores in ChromaDB. Returns chunk count to admin.
Exception	Sender not in whitelist — permission-denied message; no ingestion.
Postcondition	ChromaDB collection “rulebook” updated; all decisions grounded in new policy.

6.3 UC5 — Approve / Reject / Flag

Use Case ID	UC5
Name	Approve / Reject / Flag
Primary Actor	Decision Engine (Internal)
Precondition	All slots validated; HRIS record retrieved; RAG chunks retrieved.
Main Flow	LLM reasons over employee data, requested asset, budget cap, product catalog, and policy chunks. Outputs JSON with fields: decision, reason, suggested_alternative, policy_reference.
Alt Flow A	Out-of-stock — suggest next available model in same category.
Alt Flow B	Price exceeds budget cap — suggest cheaper model.
Alt Flow C	Duplicate request within 30 days — reject with reason.
Alt Flow D	No matching product — flag for human review.
Exception	Inactive employee — immediate rejection before LLM call.
Postcondition	Verdict persisted; confirmation or rejection card with policy reference sent.

6.4 UC6 — Detect Suspicious Input

Use Case ID	UC6
Name	Detect Suspicious Input
Primary Actor	Suspicion Scorer (Internal)
Precondition	Any message received from any user.
Scoring Rules	+80: prompt-injection regex match. +40: grade/model mismatch detected. +50: ≥ 3 requests submitted in 24 hours. +15: message outside 07:00–20:00 local time.
Action ≥ 80	Block session; log to <code>audit_events</code> .
Action ≥ 40	Flag admin via Telegram DM; request proceeds.
Action < 40	Allow; continue normal flow.
Postcondition	Score, flags, and raw message (max 200 chars) stored in <code>audit_events</code> .